

Research Notes on Federated learning

Federated learning

>> Section 1

Those parameters have to be optimized depending on the constraints of the machine learning application (e.g., available computing power, available memory, bandwidth). For instance, stochastically choosing a limited fraction C of nodes for each iteration diminishes computing cost and may prevent overfitting, in the same way that stochastic gradient descent can reduce overfitting.

>> Section 2

Transportation: self-driving cars Self-driving cars encapsulate many machine learning technologies to function: computer vision for analyzing obstacles, machine learning for adapting their pace to the environment (e.g., bumpiness of the road). Due to the potential high number of self-driving cars and the need for them to quickly respond to real world situations, traditional cloud approach may generate safety risks. Federated learning can represent a solution for limiting volume of data transfer and accelerating learning processes.

>> Section 3

Decentralized federated learning In the decentralized federated learning setting, the nodes are able to coordinate themselves to obtain the global model. This setup prevents single point failures as the model updates are exchanged only between interconnected nodes without the orchestration of the central server. Nevertheless, the specific network topology may affect the performances of the learning process. See blockchain-based federated learning and the references therein.

>> Section 4

$$f(\mathbf{x}_1, \dots, \mathbf{x}_K) = \frac{1}{K} \sum_{i=1}^K f_i(\mathbf{x}_i) \quad \{\displaystyle f(\mathbf{x}_1, \dots, \mathbf{x}_K) = \frac{1}{K} \sum_{i=1}^K f_i(\mathbf{x}_i)\}$$

>> Section 5

Definition Federated learning aims at training a machine learning algorithm, for instance deep neural networks, on multiple local datasets contained in local nodes without explicitly exchanging data samples. The general principle consists in training local models on local data samples and exchanging parameters (e.g. the weights and biases of a deep neural network) between these local nodes at some frequency to generate a global model shared by all nodes. The main difference between federated learning and distributed learning lies in the assumptions made on the properties of the local datasets, as distributed learning originally aims at parallelizing computing power where federated learning originally aims at training on

heterogeneous datasets. While distributed learning also aims at training a single model on multiple servers, a common underlying assumption is that the local datasets are independent and identically distributed (i.i.d.) and roughly have the same size. None of these hypotheses are made for federated learning; instead, the datasets are typically heterogeneous and their sizes may span several orders of magnitude. Moreover, the clients involved in federated learning may be unreliable as they are subject to more failures or drop out since they commonly rely on less powerful communication media (i.e. Wi-Fi) and battery-powered systems (i.e. smartphones and IoT devices) compared to distributed learning where nodes are typically datacenters that have powerful computational capabilities and are connected to one another with fast networks.

>> Section 6

Industry 4.0: smart manufacturing In Industry 4.0, there is a widespread adoption of machine learning techniques to improve the efficiency and effectiveness of industrial process while guaranteeing a high level of safety. Nevertheless, privacy of sensitive data for industries and manufacturing companies is of paramount importance. Federated learning algorithms can be applied to these problems as they do not disclose any sensitive data. In addition, FL also implemented for PM2.5 prediction to support Smart city sensing applications.

>> Section 7

Biometrics Federated Learning (FL) is transforming biometric recognition by enabling collaborative model training across distributed data sources while preserving privacy. By eliminating the need to share sensitive biometric templates like fingerprints, facial images, and iris scans, FL addresses privacy concerns and regulatory constraints, allowing for improved model accuracy and generalizability. It mitigates challenges of data fragmentation by leveraging scattered datasets, making it particularly effective for diverse biometric applications such as facial and iris recognition. However, FL faces challenges, including model and data heterogeneity, computational overhead, and vulnerability to security threats like inference attacks. Future directions include developing personalized FL frameworks, enhancing system efficiency, and expanding FL applications to biometric presentation attack detection (PAD) and quality assessment, fostering innovation and robust solutions in privacy-sensitive environments.

>> Section 8

Governance Governance considerations may also limit the applicability of federated learning, in particular when

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applied in cross-organisational settings. Most federated learning frameworks rely on a central coordinating server, which introduces questions regarding infrastructure control, model ownership, and decision-making authority among collaborators. In multi-party deployments, especially those spanning competitive or collaborative data markets, the absence of agreed-upon governance structures can create uncertainty about responsibilities for validating model updates, auditing system behaviour, and managing security incidents. These organisational issues may deter participants from contributing to shared models and can complicate regulatory compliance, thereby constraining adoption.

>> Section 9

Use cases Federated learning typically applies when individual actors need to train models on larger datasets than their own, but cannot afford to share the data in itself with others (e.g., for legal, strategic or economic reasons). The technology yet requires good connections between local servers and minimum computational power for each node.

>> Section 10

Hybrid federated dual coordinate ascent (HyFDCA) Very few methods for hybrid federated learning, where clients only hold subsets of both features and samples, exist. Yet, this scenario is very important in practical settings. Hybrid Federated Dual Coordinate Ascent (HyFDCA) is a novel algorithm proposed in 2024 that solves convex problems in the hybrid FL setting. This algorithm extends CoCoA, a primal-dual distributed optimization algorithm introduced by Jaggi et al. (2014) and Smith et al. (2017), to the case where both samples and features are partitioned across clients. HyFDCA claims several improvement over existing algorithms:

>> Section 11

Federated learning parameters Once the topology of the node network is chosen, one can control different parameters of the federated learning process (in addition to the machine learning model's own hyperparameters) to optimize learning:

>> Section 12

Optimizing the objective function $f(\mathbf{x}_1, \dots, \mathbf{x}_K)$ $\{\displaystyle f(\mathbf{x}_1, \dots, \mathbf{x}_K)\}$. Achieving consensus on $\mathbf{x}_i$ $\{\displaystyle \mathbf{x}_i\}$. In other words, $\mathbf{x}_1, \dots, \mathbf{x}_K$ $\{\displaystyle \mathbf{x}_1, \dots, \mathbf{x}_K\}$ converge to some common $\mathbf{x}$ $\{\displaystyle \mathbf{x}\}$ at the end of the training process.

>> Section 13

Covariate shift: local nodes may store examples that have different statistical distributions compared to other nodes. An example occurs in natural language processing datasets where people typically write the same digits/letters with different stroke widths or slants. Prior probability shift: local nodes may store labels that have different statistical distributions compared to other nodes. This can happen if datasets are regional and/or demographically partitioned. For example, datasets containing images of animals vary significantly from country to country. Concept drift (same label, different features): local nodes may share the same labels but some of them correspond to different features at different local nodes. For example, images that depict a particular object can vary according to the weather condition in which they were captured. Concept shift (same features, different labels): local nodes may share the same features but some of them correspond to different labels at different local nodes. For example, in natural language processing, the sentiment analysis may yield different sentiments even if the same text is observed. Unbalanced: the amount of data available at the local nodes may vary significantly in size. The loss in accuracy due to non-iid data can be bounded through using more sophisticated means of doing data normalization, rather than batch normalization.